

A comparative study of the predictive power of component-based approaches to structural equation modeling

Structural
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Abstract

Purpose – Generalized structured component analysis (GSCA) and partial least squares path modeling (PLSPM) are two key component-based approaches to structural equation modeling that facilitate the analysis of theoretically established models in terms of both explanation and prediction. This study aims to offer a comparative evaluation of GSCA and PLSPM in a predictive modeling framework.

Design/methodology/approach – A simulation study compares the predictive performance of GSCA and PLSPM under various simulation conditions and different prediction types of correctly specified and misspecified models.

Findings – The results suggest that GSCA with reflective composite indicators (GSCA_R) is the most versatile approach. For observed prediction, which uses the component scores to generate prediction for the indicators, GSCA_R performs slightly better than PLSPM with mode A. For operative prediction, which considers all parameter estimates to generate predictions, both methods perform equally well. GSCA with formative composite indicators and PLSPM with mode B generally lag behind the other methods.

Research limitations/implications – Future research may further assess the methods' prediction precision, considering more experimental factors with a wider range of levels, including more extreme ones.

Practical implications – When prediction is the primary study aim, researchers should generally revert to GSCA_R, considering its performance for observed and operative prediction together.



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Originality/value – This research is the first to compare the relative efficacy of GSCA and PLSPM in terms of predictive power.

Keywords Prediction, Simulation, Structural equation modeling, Generalized structured component analysis, Partial least squares path modeling

Paper type Research paper

Introduction

In many areas of marketing research and generally in business research, the concepts of interest are not directly observable and are therefore generally measured with scales comprising sets of items. These items indirectly measure the concept of interest, assuming that a certain underlying construct manifests itself through these items. A number of these constructs include individual perceptions and attitudes, such as perceived product quality (Roshan and Marcel, 2020), service quality (Praveen *et al.*, 2018) and brand commitment (Ceridwyn and Debra, 2012). Other constructs capture organizational traits, such as brand orientation (Yen-Tsung and Ya-Ting, 2013), firm-wide integrated marketing communication (Porcu *et al.*, 2017) and transparency in corporate marketing (Leitch, 2017). To analyze the complex inter-relationships between items and constructs, marketing researchers increasingly draw on component-based structural equation modeling (SEM) methods, such as generalized structured component analysis (GSCA; Hwang and Takane, 2004, 2014) and partial least square path modeling (PLSPM) (Lohmöller, 1989; Wold, 1982), which use weighted composites of indicators (components) [1] as statistical proxies for constructs (Rigdon *et al.*, 2017, 2019).

A main concern in the use of component-based SEM methods is whether the estimated models are also useful for prediction purposes, where the aim is to apply model parameters estimated from a training sample to generate falsifiable predictions for other observations not used in the model estimation (Cepeda Carrión *et al.*, 2016; Richter *et al.*, 2016). These observations can be excluded from the main sample or they can be collected at a future point in time or even in another context (Shmueli *et al.*, 2016). Establishing a model's predictive performance is fundamental for its practical relevance (Shugan, 2009) and a primary criterion for evaluating theoretical falsifiability (Popper, 1962). In fact, managerial implications – as commonly reported in marketing research articles – are inherently predictive in nature in that they suggest that the study findings generalize beyond the data and specific context at hand (Hair and Sarstedt, 2021).

While an extensive body of literature has focused on assessing component-based SEM methods in terms parameter recovery (e.g. whether their parameter estimates are unbiased; Areskoug, 1982; Cho and Choi, 2020; Goodhue *et al.*, 2006; Hwang *et al.*, 2010; Sarstedt *et al.*, 2016), evaluating their performances in terms of prediction is mostly absent in prior research. Extant studies have examined PLSPM's predictive performance in isolation (Becker *et al.*, 2013; Evermann and Tate, 2016; Shmueli *et al.*, 2016), or developed procedures and indices for evaluating a model's predictive power (Cho *et al.*, 2019; Liengaard *et al.*, 2021; Sharma *et al.*, 2021; Shmueli *et al.*, 2019). However, no study has yet systematically compared the predictive performance of GSCA and PLSPM.

Addressing this gap in research, this study empirically examines the predictive performance of the two component-based approaches under various simulation conditions, including the number of indicators per component, weight parameter values, structural model complexity, correlations between exogenous components, types of components (i.e. nomological or canonical; Cho and Choi, 2020), model specifications and sample sizes. The simulation study considers both modes of PLSPM (i.e. PLSPM with mode A and PLSPM

with mode B) and two versions of GSCA (i.e. GSCA with reflective composite indicators and GSCA with formative composite indicators) that may be considered GSCA counterparts to PLSPM's modes A and B (Cho and Choi, 2020).

The results suggest that GSCA with reflective composite indicators is the most versatile approach. For observed prediction, which uses the component scores to generate prediction for the indicators, the method performs slightly better than PLSPM with mode A. For operative prediction, which considers all parameter estimates to generate predictions, both methods perform equally well. GSCA with formative composite indicators and PLSPM with mode B generally lag behind the other methods.

Brief theoretical comparisons of generalized structured component analysis and partial least squares path modeling

As numerous publications have already discussed GSCA and PLSPM (Cho *et al.*, 2021; Hwang *et al.*, 2010; Hwang, Sarstedt, *et al.*, 2020), the following sections provide a brief comparison of the methods' key distinguishing features in terms of model specification, model estimation and model evaluation. Complementing this discussion, Table 1 summarizes the key similarities and differences between GSCA and PLSPM. Hwang and Takane (2004, Chapter 2), offer a detailed discussion of GSCA's statistical properties and Lohmöller (1989) provides a detailed description of the statistics behind PLSPM.

Model specification

While both methods rely on measurement and structural models to express the relationships between the components and indicators, GSCA additionally specifies another sub-model – the weighted relation model – that defines components as weighted sums of their associated indicators. PLSPM does not explicitly include the weighted relation model, but it considers such a sub-model implicitly because component scores in PLSPM are always computed as weighted sums of indicators during its estimation process. Another difference between GSCA and PLSPM relates to the integration of the sub-models. GSCA combines its three sub-models into a unified algebraic formulation (i.e. a single equation), referred to as the GSCA model, which facilitates the definition of a global optimization criterion. PLSPM on the other hand addresses the measurement and structural models separately, each being linked to a separate (local) optimization criterion.

Further differences exist with regard to structural and measurement model specification. GSCA facilitates the specification of nonrecursive relationships, where the structural model includes circular relationships between the components. In PLSPM, on the other hand, structural model relationships are typically considered recursive. Also, GSCA permits each indicator to be linked to multiple components (Hwang and Takane, 2014, Chapter 2), while in PLSPM, each indicator is usually assigned to only one component (Hair *et al.*, 2022, Chapter 2).

Model estimation

Model estimation in GSCA draws on the alternating least squares (ALS) algorithm, which optimizes a single criterion, derived from the GSCA model that combines the three sub-models into one algebraic formulation. The ALS algorithm converges when the difference in GSCA's optimization function decreases less than a pre-specified tolerance criterion between two consecutive iterations. In contrast, the fixed-point algorithm in PLSPM involves minimizing separate local optimization functions by solving a series of regression analyses. Convergence occurs when the indicator weights do not change substantially from one iteration to the next.

Table 1.
Similarities and
differences between
GSCA and PLSPM

<i>Model specification</i>	GSCA	PLSPM
Sub-models	• Measurement model • Structural model • Weighted relation model	• Measurement model • Structural model
Models integrated into a single equation	Yes	No
Reciprocal relationships between components allowed	Yes	No ⁺
Indicator types	• Reflective composite indicators • Formative composite indicators	• Reflective composite indicators • Formative composite indicators
Indicators can be assigned to multiple components	Yes	No ⁺
<i>Model estimation</i>		
Algorithm type	Alternating least squares	Fixed-point least squares
Global optimization criterion	Yes	No
Normality assumption	No	No
Convergence	Change in optimization function between two consecutive iterations	Change in indicator weights between two consecutive iterations
<i>Model evaluation</i>		
Local model fit	Yes	Yes
Global model fit	Yes	(No) ⁺⁺
Model comparisons	Yes	Yes
Statistical inference	Yes, via resampling methods	Yes, via resampling methods

Notes: [†]Extensions of the original PLSPM algorithm overcome these limitations in model specification (Lohmöller, 1989, Chapter 2). ⁺⁺ Although PLSPM does not follow any global optimization criterion, some researchers use global model fit criteria such as SRMR and NFI, which do not fit the method properly (Chin *et al.*, 2020; Lohmöller, 1989, Chapter 2; Hair *et al.*, 2019)

Model evaluation

While both methods rely on a series of measures to assess the component measures' reliability and validity (Franke and Sarstedt, 2019; Hair, Risher, *et al.*, 2019; Henseler *et al.*, 2015; Hwang and Takane, 2014) as well as the model's predictive power (Cho *et al.*, 2019; Shmueli *et al.*, 2016), GSCA also facilitates the computation of overall model fit measures, which evaluate how well a model fits the data. Research has suggested measures, such as the goodness-of-fit index and the standardized root mean square residual, which quantify the discrepancies between the sample and model-implied covariances (Cho *et al.*, 2020). The concept of model fit as defined in GSCA is not applicable to PLSPM, as the method does not consider a global optimization function. Nevertheless, research has proposed several PLSPM-based model fit measures, which, however, in their current form, offer limited potential to identify model misspecifications (Chin *et al.*, 2020; Hair, Sarstedt, *et al.*, 2019; Hair *et al.*, 2020).

Simulation study

Simulation design

The simulation study sets out to assess and compare the predictive performance of GSCA and PLSPM, taking into consideration a broad range of simulation conditions. The study considered two modes of PLSPM: mode A (PLS_A) and mode B (PLS_B). To be consistent with the two modes of PLSPM, the simulation study considered two separate GSCA methods: GSCA with reflective composite indicators (GSCA_R) and GSCA with formative composite indicators (GSCA_F) (Cho and Choi, 2020). In GSCA_R, all indicators are specified as reflective composite indicators, where the component explains the variances of its associated indicators (and dependent components). On the contrary, in GSCA_F, all indicators are formative composite indicators, where the component explains the variances of its dependent components only. Irrespective of whether using GSCA_R or GSCA_F, the method always linearly combines the indicators to form components to represent the concepts of interest (hence reflective and formative *composite* indicators). Therefore, "variation in the indicators precedes variation in the latent variable" (Borsboom *et al.*, 2003, p. 208), which – from a measurement theory perspective – means that, by definition, components are inextricably tied to their measures (Diamantopoulos, 2006). When GSCA_F is applied, loadings are estimated in the same manner as the second stage of PLSPM after components are estimated via the ALS algorithm.

In the study, various experimental factors were manipulated, which have frequently been considered in SEM simulation research (Bandalos and Gagné, 2012; Paxton *et al.*, 2001) and are likely affect the methods' performances (Cho *et al.*, 2021; Cho and Choi, 2020; Hair *et al.*, 2017; Hwang *et al.*, 2010). These factors included the number of indicators per component, the ratio of weight differences, structural model complexity, correlations between exogenous components, component type, model specification and sample size. The levels of each experimental factor correspond to those reported in prior simulation studies on related topics (Cho and Choi, 2020; Cole and Preacher, 2014; Hair *et al.*, 2017; Hwang, Cho, *et al.*, 2021; Sarstedt *et al.*, 2016).

Specifically, the study considered three different numbers of indicators per component in the measurement model ($J_p = 3, 6$ and 9). Once the number of indicators per component (J_p) is selected, each component's indicators were divided into three groups of the same size, for example, $J_p = 6$ for each component resulted in three groups of two indicators each. The same weight values were assigned to indicators in the same group. Then, the ratio of weight differences between groups was varied across three levels ($dif_w = 0.1, 0.2$ and 0.3). The ratio of weight differences was manipulated instead

of their absolute size differences, because the number of indicators per component impacts the weight sizes (i.e. the more indicators per component, the smaller weights for the indicators on average). Thus, the same weight difference can have differential influences depending on the number of indicators per component. For example, a weight difference of 0.1 may be considered relatively large for small weights (e.g. about 0.2 or smaller), while relatively small for large weights (e.g. about 0.5 or larger). Accordingly, it is more reasonable to control the ratio of weight differences instead of their absolute size differences when the number of indicators per component varies. As a result, the study considered nine measurement model setups (i.e. three different numbers of indicators times three weight difference ratios).

Moreover, the study considered three levels of structural model complexity (i.e. simple, intermediate, complex). As displayed in [Figure 1](#), a more complex model has a larger number of components, path coefficients and non-zero correlations between exogenous components. For example, the simple structural model involves four components, three path coefficients and one exogenous component correlation, whereas the complex structural model includes seven components, ten path coefficients and three exogenous component correlations. The study took two levels of exogenous component correlation ($r = 0.1$ and 0.3) for each structural model into account, which resulted in six structural models (i.e. three structural complexities times two exogenous component correlations).

Furthermore, the study considered two types of components (i.e. nomological and canonical components; [Cho and Choi, 2020](#)). Nomological components explain the variances of indicators and dependent components in the measurement and structural models to the greatest extent possible (for an in-depth explanation of these components, see [Cho and Choi, 2020](#)). Canonical components maximize only the explained variances of dependent components in the structural model as in canonical correlation analysis ([Cho and Choi, 2020](#); [Dijkstra, 2017](#); [Tenenhaus et al., 2017](#)). From a prediction perspective, nomological components are designed for the prediction of indicators and dependent components, whereas canonical components focus solely on the prediction of dependent components.

In total, the study design yielded 108 different data generation scenarios in a full factorial design (i.e. nine different measurement models times six different structural models times two component types). For each scenario, two model specifications were considered: the correct, data generating model and a misspecified model that additionally included irrelevant path coefficients, as depicted by dotted lines in [Figure 1](#).

Based on [Cho and Choi's \(2020\)](#) procedure, the implied correlation matrix was obtained for each of the different data generating scenarios. For each scenario, 1,000 training sets of four different sample sizes ($N = 100, 200, 500$ and $1,000$) were generated for parameter estimation and one test set of $N = 10,000$ for the evaluation of the methods' predictive performance. For the four different sample sizes of each training set, 432,000 training sets (i.e. 108 data generating scenarios times four sample sizes times 1,000 training sets) and 108 test sets were created. The study uses MATLAB codes for both GSCA and PLSPM methods, developed by the authors.

Measures of predictive performance

The predictive performance of the methods (i.e. $GSCA_R$, $GSCA_F$, PLS_A and PLS_B) was evaluated in terms of two types of prediction: *observed* and *operative*. Observed prediction refers to a prediction of the indicators via their components defined as:

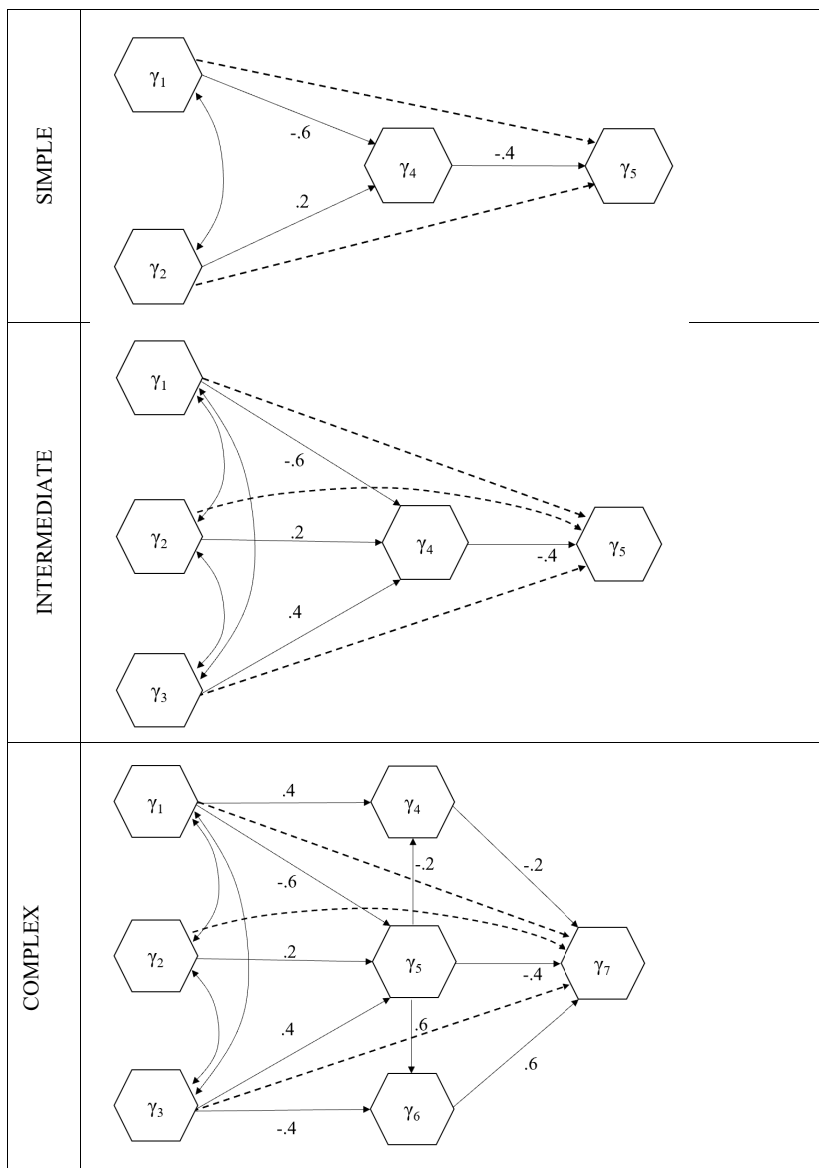


Figure 1. Models used for data generation

Notes: The hexagons represent components. The straight lines denote the true paths, whereas the dashed lines are the false paths or the paths of zero values

$$\hat{\mathbf{Z}} = \mathbf{Z}\hat{\mathbf{W}}\hat{\mathbf{C}}. \quad (1)$$

where $\mathbf{Z}$ is a matrix of indicators, $\hat{\mathbf{Z}}$ is a matrix of indicators' predicted values, $\hat{\mathbf{W}}$ is a matrix of weight estimates and $\hat{\mathbf{C}}$ is a matrix of loading estimates. In many studies using GSCA or PLSPM (Rego *et al.*, 2013; Romdhani *et al.*, 2015), where all observed variables considered are indicators of components, the observed prediction's accuracy suggests how well components are constructed to predict their indicators.

Operative prediction (Lohmöller, 1989) is a prediction of the dependent components' indicators based on their independent components' indicators:

$$\hat{\mathbf{Z}} = \mathbf{Z}\hat{\mathbf{W}}\hat{\mathbf{B}}\hat{\mathbf{C}}, \quad (2)$$

where $\hat{\mathbf{B}}$ is the matrix of estimated path coefficients. This operative prediction involves all parameter estimates. Although other types of prediction can also be considered theoretically (Lohmöller, 1989; Shmueli *et al.*, 2016), this study focused on observed and operative prediction because only in these types of prediction both predictors and predictands (variables to be predicted) are indicators that have data. Thus, the computation of the predicted values and the evaluation of their predictive accuracy are always possible.

For each of the two prediction types, an out-of-sample prediction error was computed, which allows to assess a model's predictive accuracy on a test sample that was not used for parameter estimation. The quantification of the out-of-sample prediction errors relies on the mean squares error (MSE), which has become a standard measure for assessing a method's predictive performance (Chica and Rand, 2017; Nau, 2016). Specifically, the MSEs of the two prediction types considered in the study are computed as follows:

$$SS(\mathbf{Z}_{test} - \mathbf{Z}_{test}\hat{\mathbf{W}}\hat{\mathbf{C}}) \text{ for observed prediction} \quad (3)$$

$$SS(\mathbf{Z}_{test} - \mathbf{Z}_{test}\hat{\mathbf{W}}\hat{\mathbf{B}}\hat{\mathbf{C}}) \text{ for operative prediction,} \quad (4)$$

where $SS()$ stands for sum of squares, $\mathbf{Z}_{test}$ is a matrix of indicators in a test sample and $\hat{\mathbf{W}}$, $\hat{\mathbf{C}}$ and $\hat{\mathbf{B}}$ are matrices of parameter estimates obtained from a training sample. All indicators in $\mathbf{Z}_{test}$ and components of $\mathbf{Z}_{test}$ based on $\hat{\mathbf{W}}$ in equations (3) and (4) should be standardized to have zero means and unit variances in advance.

Both (squared) bias and variance of a method contribute to the MSE. There is a well-known property between these two quantities, called the bias-variance tradeoff, that a method's variance can be reduced by increasing its bias (Hastie *et al.*, 2001, pp. 37–38). In some cases, hence, a method with no bias may have a larger variance, resulting in a larger MSE, than another method that has some bias yet a much smaller variance. The aim is to find a method that achieves a good balance between bias and variance, as evidenced in small MSE values. A method with a smaller MSE shows a better predictive performance (i.e. generalizes better to unseen samples).

Results

The assessment of the methods' overall relative performances across the different experimental conditions draws on their average MSE values per experimental factor, computed by aggregating the individual MSE values over the other factors. The following descriptions focus on reporting the results under the data generation (correct) models because those under the misspecified models were quite comparable (see Figures A1 and A2

in the Appendix). The next sections document the methods' overall performances for observed and operative prediction of the correctly specified models, and, more briefly, for the misspecified models.

Performance for observed prediction

Figure 2 displays the methods' average MSE values and their 95% confidence intervals across the conditions of each experimental factor in observed prediction under the models with nomological and canonical components. Morel's (2018) MATLAB data visualization toolbox, Gramm, was used to plot the 95% confidence intervals. As shown in the figure, in general, $GSCA_R$ shows the best performance in observed prediction, followed by PLS_A , $GSCA_F$ and PLS_B . The results also show that the performance differences among the methods are not substantially large and exhibit similar patterns across all the conditions. The average MSE values of the methods also exhibit quite similar patterns between the models with nomological and canonical components across the experimental factors, suggesting that the component type has no substantial bearing on the methods' observed predictive performance. However, the number of indicators tends to affect the methods' observed prediction negatively (i.e. the more indicators per component, the larger the MSE values), suggesting that all the methods perform better in observed prediction under simpler measurement models with a smaller number of indicators.

When the number of indicators increases, the MSE deviations between the pair of $GSCA_F$ and PLS_B and the pair of $GSCA_R$ and PLS_A become larger, whereas the MSE deviations within the same pair remain the same. Moreover, the ratio of indicator weight differences tends to influence the methods' observed prediction positively (i.e. the larger the ratio, the smaller MSE values). This result indicates that the methods tend to better predict indicators' unseen values through their components when the indicators contribute more differentially to forming their components. Sample size tends to have a positive impact on $GSCA_F$'s and PLS_B 's observed prediction, whereas this factor does not seem to affect $GSCA_R$'s and PLS_A 's observed prediction. When sample size is large ($N = 1,000$), $GSCA_F$ appears to perform similarly to $GSCA_R$ and PLS_A .

The other experimental factors (structural model complexity and exogenous component correlation) do not seem to have substantial effects on the methods' observed prediction. Nonetheless, PLS_A and PLS_B tend to perform slightly worse when the structural model is simple, and the exogenous component correlation is 0.3.

Performance for operative prediction

Figure 3 shows the methods' average MSE values and their 95% confidence intervals per experimental factor in operative prediction under the models with nomological and canonical components. On average, there are little MSE differences between the pair of $GSCA_R$ and PLS_A and between the pair of $GSCA_F$ and PLS_B in all conditions, indicating that each pair performs virtually identically in predicting unseen values of dependent components' indicators using their independent components' indicators. This leads the MSE differences between the pairs to be more salient. Thus, the following descriptions focus on each experimental factor's influence on the operative predictive performance of the pairs rather than that of the individual methods. As in the observed prediction case, the methods' MSE patterns per experimental factor largely remain the same between the models with nomological and canonical components. However, the MSE differences themselves between the pairs appear to be somewhat smaller under the models with canonical components.

In general, the number of indicators tends to influence the pairs' operative prediction negatively, indicating that each pair shows better performance in operative prediction

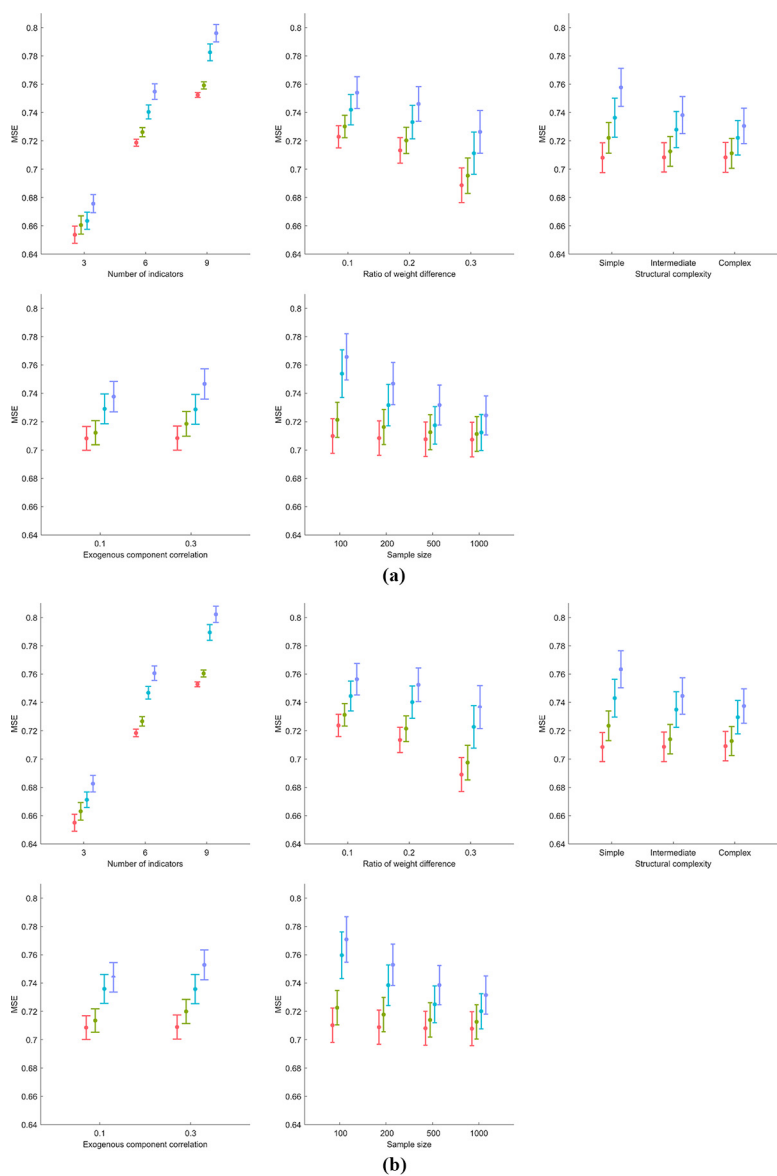
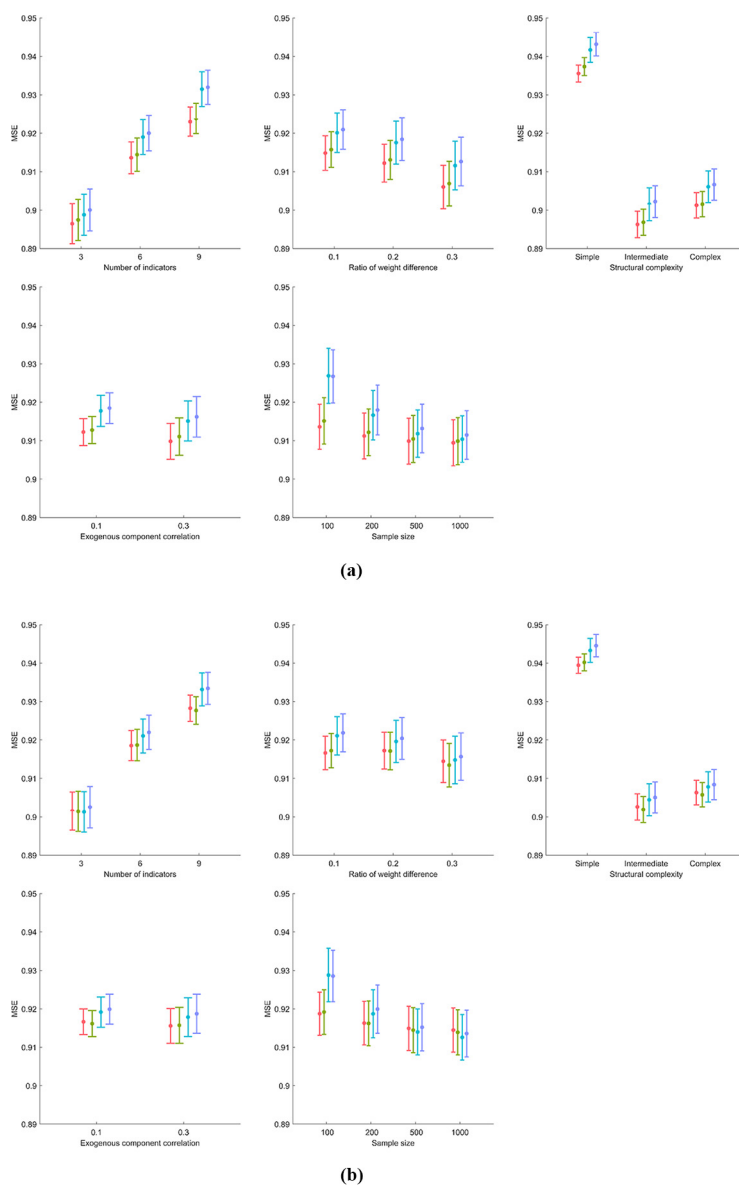


Figure 2.
Performance for
observed prediction

Notes: (a) Nomological components; (b) canonical components. The average MSE values and their 95% confidence intervals of $GSCA_R$, PLS_A , $GSCA_F$, and PLS_B per condition under the correct specification of two data generating models with nomological or canonical components for observed prediction (red: $GSCA_R$, green: PLS_A , blue: $GSCA_F$, purple: PLS_B)



Notes: (a) Nomological components; (b) canonical components. The average MSE values and their 95% confidence intervals of $GSCA_R$, PLS_A , $GSCA_F$, and PLS_B per condition under the correct specification of two data generating models with nomological or canonical components for operative prediction (red: $GSCA_R$, green: PLS_A , blue: $GSCA_F$, purple: PLS_B)

Figure 3.
Performance for
operative prediction

when the number of indicators becomes smaller. When the number of indicators is equal to three, the pairs perform quite comparably to each other, particularly under the models with canonical components. Moreover, the ratio of weight differences tends to affect the pairs' operative prediction positively, indicating that each pair is likely to perform better when the indicators' weights are different to a greater degree. The pairs seem to perform quite similarly when the weight difference ratio is relatively large ($dif_w = 0.3$) under the models with canonical components. Also, structural model complexity generally influences the pairs' operative prediction. Both pairs show the best operative predictive performance when the structural model's complexity is intermediate or complex. Sample size tends to have a positive influence on the pair of $GSCA_F$ and PLS_B only. Thus, with larger sample sizes (e.g. $N = 1,000$), both pairs appear to perform quite similarly. Conversely, on average, exogenous component correlation does not influence the pairs' operative prediction substantially.

Performance in model misspecification

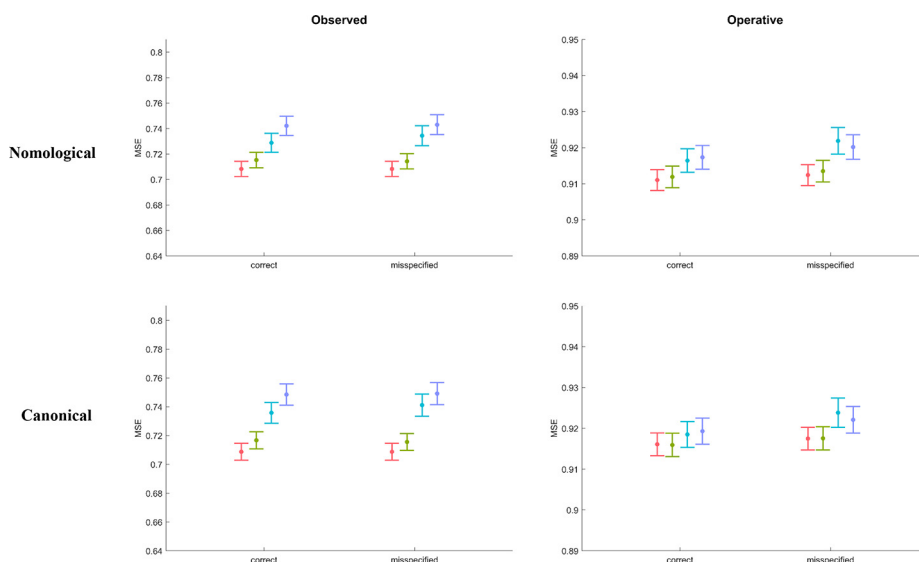
The final analysis compares the SEM methods' relative predictive performances in both types of prediction between the correct and the misspecified models with nomological and canonical components. Figure 4 displays their average MSE values and their 95% confidence intervals under the two model specifications.

The results show that in general, the methods' relative predictive performances in both types of prediction differ only slightly between the correct and misspecified models. Specifically, model misspecification hardly influences the methods' observed prediction under both nomological and canonical component models. Only $GSCA_F$ tends to perform slightly worse when the model was misspecified. Similarly, model misspecification does not seem to affect $GSCA_R$'s and PLS_A 's operative prediction substantially under the nomological and canonical component models. It tends to influence only $GSCA_F$'s and PLS_B 's operative prediction marginally in a negative fashion. For this reason, the results under the misspecified models are not presented in detail, as noted earlier. The previously presented results are transferable to the case of the misspecified models.

The methods' similar performance under correctly and misspecified models might appear surprising at first sight. However, this finding is grounded in the conceptual differences between explanation and prediction. In explanation, the focus is on minimizing bias to obtain the most accurate representation of the data. In contrast, predictive model estimation seeks to minimize the combination of bias and variance, "occasionally sacrificing theoretical accuracy for improved empirical precision" (Shmueli, 2010, p. 6). Hence, a misspecified model may achieve a better predictive performance than a correctly specified model (Chin *et al.*, 2020).

Discussion

Assessing a model's predictive power is at the heart of a scientific enterprise. Making falsifiable predictions about new observations allows marketing researchers to evaluate and compare existing theories. Marketing researchers routinely derive practical prescriptions, as characteristically documented in "Managerial Implications" sections, which inherently follow a predictive perspective (Sarstedt and Danks, 2021). For example, marketing researchers frequently make conditional statements that foreshadow a specific result if a specific activity is implemented (i.e. prescriptive statements) such as Cuong *et al.* (2020, p. 2274) when concluding that "companies should devote their attention to strengthening their brand equity because of its directly positive mediating effect of the nexus between relationship equity and brand loyalty." Considering the importance of prediction for the



Notes: The average MSE values and their 95% confidence intervals of GSCA_R, PLS_A, GSCA_F, and PLS_B per condition under two model specifications of two data generating models with nomological or canonical components (red: GSCA_R, green: PLS_A, blue: GSCA_F, purple: PLS_B)

Figure 4.
The average MSE values and their 95% confidence intervals of the correct and misspecified models

marketing, it is surprising that prior research has not systematically evaluated the relative performance of component-based SEM methods, which are particularly useful for prediction in the confinement of a model derived from theory and logic.

Addressing this gap in research, this study reports the results of an extensive simulation study that examines the relative performance of GSCA and PLSPM for two types of prediction (i.e. observed and operative) under different model and data generation set-ups (i.e. structural model complexity, number of indicators, weight difference, sample size, model specification, exogenous component correlation and component type). The simulation study shows that GSCA_R generally performs best for predicting indicators' unseen values through their components (i.e. observed prediction), although its performance appears to be marginally better than that of PLS_A. An interesting finding is that GSCA_R and PLS_A provide a better observed prediction than GSCA_F and PLS_B even under models with canonical components, where these methods are known to produce biased estimates of weights and loadings (Cho and Choi, 2020). This result supports that biased parameter estimates do not necessarily lead to lower predictive power (Hastie *et al.*, 2001, pp. 37–38).

The results also show that GSCA_R and PLS_A perform virtually identically for predicting unseen values of dependent components' indicators using their independent components' indicators (i.e. operative prediction). This is also the case for the pair of GSCA_F and PLS_B. Overall, GSCA_R and PLS_A consistently perform better in operative prediction than GSCA_F and PLS_B. Exceptions are situations with large sample sizes, or when considering models with canonical components, small numbers of indicators or large differences of weight ratios. In these constellations, GSCA_F and PLS_B perform equally well. Again, the pair of GSCA_R and PLS_A generally provides a better operative prediction than the pair of GSCA_F

and PLS_B even under canonical component models. Also, model misspecification does not lead to substantial differences in the methods' operative prediction.

In summary, the findings suggest that GSCA_R is the most versatile method for prediction in component-based SEM, closely followed by PLS_A. GSCA_F and PLS_B generally lag behind, matching the predictive performance of GSCA_R and PLS_A only under specific conditions, such as for operative prediction when the sample size is large (e.g. $N = 1,000$). Table 2 summarizes the methods' performance for combinations of prediction (i.e. observed and operative) and component model types (i.e. nomological and canonical).

Researchers wishing concrete guidance in their method choice are likely going to wonder whether observed or operative prediction is more relevant. From a statistical perspective, operative prediction should be taken as a standard to judge the methods' overall predictability because it takes into account all parameter estimates, as discussed earlier. Accordingly, based on their performance under operative prediction (Table 2), researchers should draw on either GSCA_R or PLS_A. This recommendation mirrors the various guidelines for using Out-of-bag Prediction Error (OPE) for GSCA (Cho *et al.*, 2019) or Q^2 for PLSPM (Chin, 1998), which have been proposed to assess a model's predictive power. For GSCA, researchers may draw on two local cross-validation indexes (OPE_M and OPE_S), which aim to assess a model's predictive power in observed prediction (i.e. predicting indicators through their components) and latent prediction (i.e. predicting dependent components based on independent components' indicators), respectively. However, OPE has been recommended as the primary criterion for model comparison in terms of predictive generalizability, as it evaluates the entire model's predictive accuracy. Likewise, when using Q^2 for PLSPM, researchers can revert to the cross-validated communality measure, which resembles observed prediction, and the cross-validated redundancy measure, which resembles operative prediction (Chin, 1998). Due to Q^2 's reliance on the structural model setup, numerous researchers suggest routinely using the cross-validated redundancy measure (Hair *et al.*, 2017; Tenenhaus *et al.*, 2005). For example, Chin (2010, p. 680) notes that "one would use the cross-validated redundancy measure to examine the predictive relevance of one's theoretical/structural model."

It is worth noting that these recommendations are entirely based on the methods' predictive power. However, predictive power is only one of many aspects to consider in method choice. Specifically, while GSCA_R and PLS_A perform equally well under operative prediction, GSCA_R offers researchers more flexibility in terms of model specification and evaluation (Table 1). For example, different from PLS_A, GSCA_R facilitates the computation of various model fit measures that allow researchers to assess whether their predictive model also has a satisfactory level of fit. This characteristic is particularly important considering that high predictive power does not necessarily go hand in hand with

Table 2.
Guidelines for
method choice based
on predictive
performance

	Observed prediction		Operative prediction	
	Nomological components	Canonical components	Nomological components	Canonical components
GSCA _R	●	●	●	●
GSCA _F	○	○	●	○
PLS _A	●	●	●	●
PLS _B	○	○	●	○
Notes: ○ = very weak performance; ● = weak performance; ● = fair performance; ● = good performance; ● = very good performance				

satisfactory model fit. Hence, $GSCA_R$ seems to be the more sensible choice when estimating models with an operative prediction focus.

At the same time, however, method choice should also consider their efficiency and robustness of parameter recovery (Cho and Choi, 2020; Hair *et al.*, 2017) as there is no dichotomy between explanatory and predictive modeling (Hofman *et al.*, 2017). Instead of exclusively subscribing to one modeling purpose only, researchers should follow a hybrid approach in which they first establish their research's main goal (e.g. explanation) and evaluate the model's performance in terms of this goal (i.e. explanatory power) but also in terms of the other goal (i.e. predictive power). Complementing explanatory and predictive modeling perspectives increases a study's managerial relevance, since "it remains true that if we can predict successfully on the basis of a certain explanation, we have a good reason, and perhaps the best sort of reason, for accepting the explanation" (Kaplan, 1964). Furthermore, this practice can guide the comparison of competing models derived from different theories by selecting a model that generalizes well to other samples, while avoiding models that overfit the data by making use of spurious sample-specific patterns (Shmueli and Koppius, 2011). If no theories are available for specifying alternative (more parsimonious) models, researchers may consider a statistical approach that combines lasso-type or $L1$ regularization into component-based SEM in order to generate such models in an automatic fashion. $GSCA$ has already been combined with $L1$ regularization (Hwang and Takane, 2014, Chapter 9).

Considering the above, further research is needed to better grasp potential tradeoffs between the methods' performance in terms of prediction and model fit. For example, researchers should evaluate under which conditions a certain level of overfit is acceptable to warrant sufficient model fit. In doing so, future studies may consider more complex model constellations, taking into account multidimensional components, non-linear relationships among components (i.e. interactions effects of components) and conditional process models (Sarstedt *et al.*, 2020). Relatedly, although the present study has considered a number of experimental factors with various levels, which have been frequently encountered in the literature, the study's scope can be further expanded by considering a wider range of factor levels, including more extreme ones to enhance the results' generalizability.

Finally, this study investigated the comparative quality of point predictions only but did not consider prediction intervals. A prediction interval can be an indicator of prediction precision, mirroring the uncertainty of prediction. In the study, the SEM approaches computed the predicted value (i.e. a point prediction) for each observation of a dependent variable and the sum of prediction errors. However, resampling methods such as bootstrapping return multiple predicted values for an observation of a dependent variable, which allow to derive the prediction interval (see more details in Shmueli *et al.*, 2016). Future research may evaluate such predictive accuracy of component-based SEM approaches.

Note

1. In this study, the terms component and composite are used interchangeably.

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Further reading

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Appendix

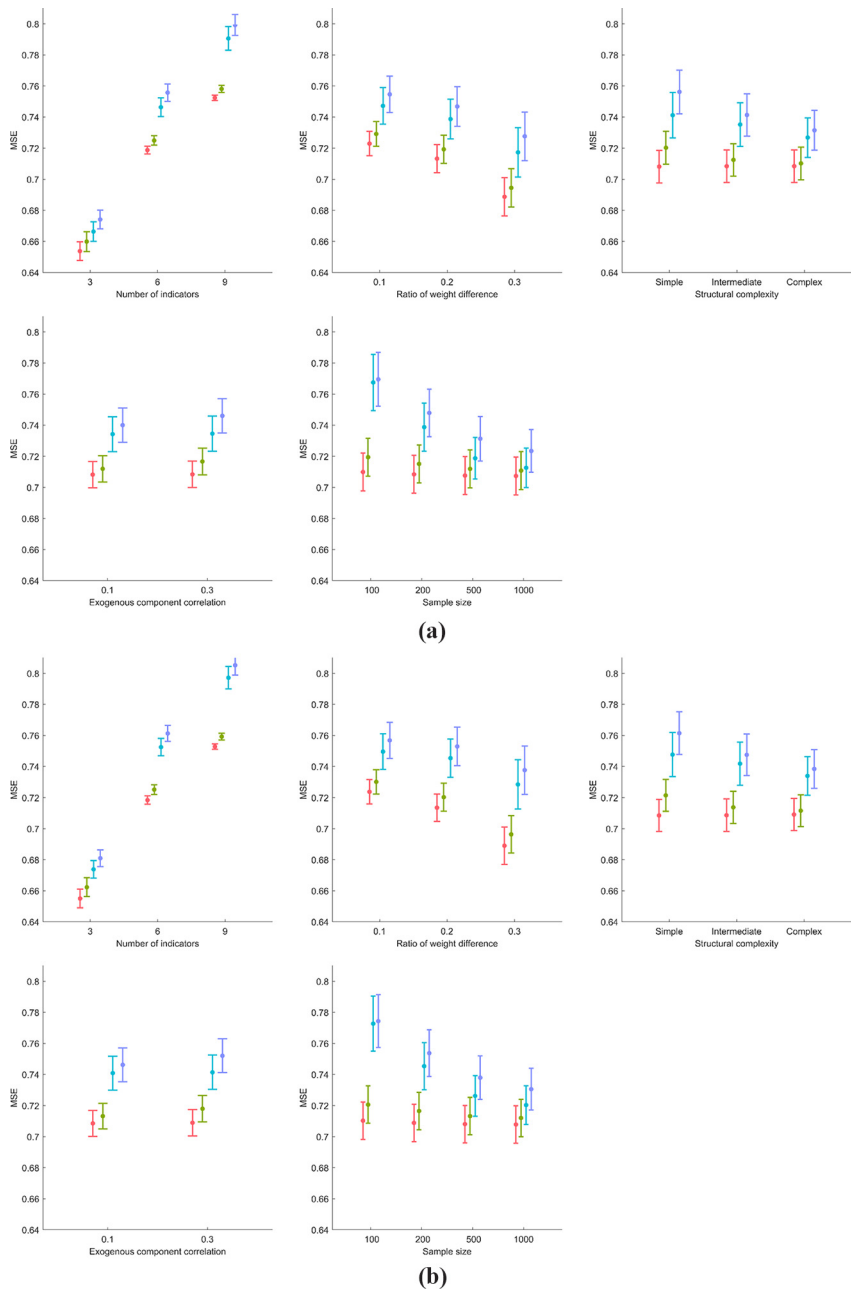
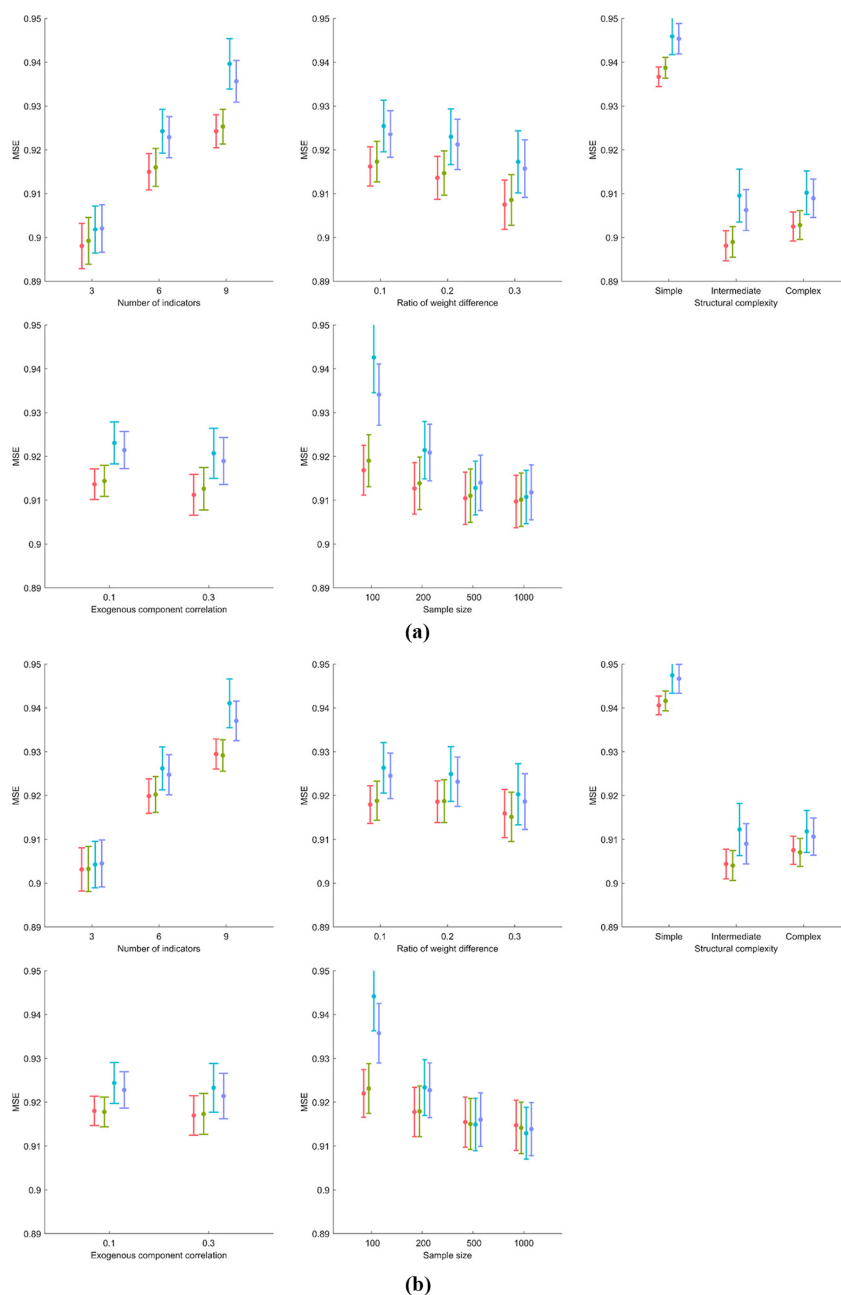


Figure A1.
Performance for
observed prediction
under model
misspecification

Notes: (a) Nomological components; (b) canonical components



Notes: (a) Nomological components; (b) canonical components

Figure A2.
Performance for
operative prediction
under model
misspecification